NOT MEASUREMENT SENSITIVE

A-A-59782
27 September 2005

COMMERCIAL ITEM DESCRIPTION

HEAT EXCHANGER CLEANING COMPOUND

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This CID covers a liquid form heat exchanger cleaning compound, hereinafter referred to as "compound", intended for removing scale deposits from heat exchangers and distilling units.

2. SALIENT CHARACTERISTICS.

2.1 Materials. Unless otherwise specified herein, materials shall be in accordance with the manufacturer's material specifications. The compound shall be a uniform liquid or be able to be made uniform with mild mixing.

2.1.1 Hydrogen-ion concentration (pH). The compound shall be an acidic solution such that the solution, when prepared for use according to manufacturer's recommendations, shall have a pH less than 2.0 when determined in accordance with ASTM E70.

2.1.2 Acid content. The compound, as prepared for use, shall contain not more than 10% by weight hydrochloric acid (HCl) or 15% by weight orthophosphoric acid (H_3PO_4) when determined in accordance with the manufacturer's standard practice.

2.1.3 Excluded ingredients. The compound shall not contain sulfuric acid or organic solvents.

2.1.4 Toxins, carcinogens, and reproductive stressors. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Conformance shall be determined by a health hazard assessment (HHA) in accordance with 2.1.4.1. Products meeting this requirement will be issued a letter of approval by the Naval Sea Systems Command. Letters of approval shall apply solely to those products specifically evaluated for toxicity. The manufacturer is required to notify the Qualifying Activity when there have been formulation changes to the qualified product. Upon approval of any reformulation, a new letter of approval shall be issued. Failure to report formulation changes may be cause for revocation of all letters of approval for the specific manufacturer and/or permanent banning of the product. Approval letters will be in effect for a maximum of three years, after which, reissue may be requested for the same formulation as the original letter of approval. Manufacturers shall provide a copy of the approval letter applicable to offered products to the contracting activity before the product will be accepted.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navsea.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

FSC 6850

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:26Z
Check the source to verify that this is the current version before use.

2.1.4.1 Health Hazard Assessment. The manufacturer shall provide sufficient information to permit a toxicological evaluation of the product to the Navy Environmental Health Center (NEHC), accompanied with a request to perform an HHA. As a minimum, the information shall include name, formula and approximate percentage by weight of each ingredient in the product; identification of its pyrolysis products; and any other such information as may be needed to permit an accurate appraisal of any toxicity problem associated with the handling, storage, application, use, removal, disposal, or combustion of the product. The request for a HHA shall also include a product material safety data sheet and a copy of the product label. The manufacturer shall provide any other information requested by NEHC to perform their evaluation. Proprietary data, which is clearly marked as such, will be held in the strictest confidence by NEHC. Any questions regarding toxicity, information required to conduct a HHA, and requests for a HHA shall be addressed to the Commanding Officer, Navy Environmental Health Center, ATTN: Hazardous Materials Department, Industrial Hygiene Directorate, 620 John Paul Jones Circle, Suite 1100, Portsmouth, VA 23708-2103. Upon receipt of the HHA, the manufacturer shall provide a copy to Commander, Naval Sea Systems Command, ATTN: SEA 05Z9, 1333 Isaac Hull Ave, SE, Stop 5123, Washington Navy Yard, DC 20376-5123.

2.1.4.2 Material Safety Data Sheet (MSDS). The contracting activity shall be provided a material safety data sheet with each contract bid-proposal or prior to contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313 and 29 CFR 1910.1200, Hazardous Communication Standard. The MSDS shall also be included with each shipment of the compound.

2.2 Scale dissolving ability. The compound, at usage concentration, shall dissolve calcium carbonate scale at a rate of one gram of solid calcium carbonate (chalk or clam shell) in 50 milliliters of compound at 25 °C in less than one hour.

2.3 Deleterious effects on non-metallic components. The compound shall not cause degradation of the non-metallic components listed in Table I. In order to determine the effects of exposure to the candidate compound, the materials listed in Table I shall be exposed to the compound at usage concentration for a period of 70 hours. The following properties shall be examined:

2.3.1 Hardness change. Durometer and/or Rockwell hardness tests (ASTM D2240 and ASTM D785, respectively) shall be performed on non-metallic specimens before and after exposure to the candidate compound. Maximum change in hardness shall be +/-5 points.

2.3.2 Permanent volume change. The change in volume of the non-metallic specimens following exposure to the compound shall be determined using ASTM D471 with the exception that the volume following exposure shall be determined after the specimen has been allowed to equilibrate at ambient conditions for 24 hours. The change in volume determined in this manner shall not exceed +/-3.5%.

2.3.3 Other visual effects. No cracking of the surface when viewed through a 7-power magnifying glass shall be permitted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:26Z
Check the source to verify that this is the current version before use.

TABLE I. Non-metallic materials list.

MATERIAL	SPECIFICATION	TITLE	NATIONAL STOCK NUMBER
Synthetic Rubber (Acrylonitrile-butadiene, NBR, 70-hardness)	SAE-AMS-R-6855 Class 1, Grade 70	Elastomer, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification For	5330-01-313-3583
Fluorocarbon Rubber (FKM, 75-hardness)	SAE-AMS-7276	Rings, Sealing, Fluorocarbon (FKM) Rubber High-Temperature-Fluid Resistant Low Compression Set 70 to 80	5331-01-005-0543
Tetrafluoroethylene Resin (TFE)	SAE-AS-8791	Hydraulic and Pneumatic Retainers (Back-Up Rings), Polytetrafluoroethylene (PTFE) Resin	5330-00-171-5779
Nitrile Rubber (Acrylonitrile-butadiene, NBR, 88-hardness)	SAE-AMS-P-5516 Class A, Low Flexibility	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, 160 °F	5330-00-260-9649
Polychloro-trifluoroethylene (PCTFE)	SAE-AMS-3650	Rods, Sheets, and Molded Shapes, Polychlorotrifluoroethylene (PCTFE) Unplasticized	4820-01-329-5682
Nylon 101 (Polyamide Type 6/6)		Commercial: DuPont Zytel 101, Polymer Corporation Nylatron GS or equivalent	9390-01-113-8890
Delrin AF		Commercial: Polymer Corporation DE 588 AS or equivalent	5365-01-497-5732

2.4 Corrosive effect on copper alloys. The compound, at usage concentration, shall have a corrosive effect on copper alloys (UNS C70600, UNS C71500 and UNS C92200) of not more than 20 mils per year (mpy) when determined using ASTM G31 for a 6-hour test at 35 °C.

2.5 Effect on protective films. Any candidate compound must not significantly alter the protective film kinetics of copper alloys, i.e., the compound shall not interfere with the formation of the protective oxide layer that naturally develops on copper alloys exposed to clean natural seawater. In order to quantify/determine the effect(s) of a candidate compound, the following must be performed:

2.5.1 Specimen preparation. A minimum of seven UNS C70600 (70/30) copper alloy specimens, (1 unconditioned, 3 to be cleaned with the candidate compound and 3 controls) each with a minimum total surface area of 17.5 square centimeters shall be prepared by polishing all exposed surfaces of the specimen first with 240 grit abrasive paper followed by 400 grit abrasive paper to remove surface films (including existing protective layers) and expose a clean metallic surface. No discoloration of the surface shall be visible following polishing. The specimens shall be rinsed in clean deionized water to remove abrasive and abraded particles.

2.5.2 Baseline curve. A potentiodynamic polarization curve shall be obtained for one freshly prepared specimen exposed to seawater for the purpose of the test (not a conditioned or control specimen). The seawater shall be from the same source as the exposure tests in 2.5.3. The specimen shall be allowed to come to equilibrium in non-flowing seawater for a minimum of 30 minutes, but no more than 60 minutes prior to initiation of the scan. An anodic polarization curve shall be generated at a scan rate of 0.6 V/hr from the corrosion potential (Ecorr) to +0.2 V against a saturated calomel electrode (SCE) for this and all subsequent polarization scans. Tests shall be in accordance with ASTM G61 with the following modifications:

a. Where 3.56% sodium chloride solution is called for in ASTM G61, clean natural seawater shall be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:26Z
Check the source to verify that this is the current version before use.

b. Do not purge the solution to remove oxygen.

c. Do not perform the reverse polarization scan.

2.5.3 Initial seawater exposure. The remaining six 70/30 copper alloy specimens prepared in 2.5.1 shall be exposed to continuously refreshed, once-through (not recirculated) natural seawater for a minimum of three weeks (21 days). Following the initial minimum 21-day exposure, potentiodynamic polarization curves must be obtained for the control specimens as in paragraph 2.5.2.

2.5.4 Control specimen criteria. The anodic current density over the range of cupronickel potentials shall decrease by a minimum of two orders of magnitude from the specimen tested in 2.5.2. If this criterion is not met for the control specimens, the test must be repeated starting at 2.5.1.

2.5.5 Test specimens. Following the initial 21-day exposure, three test specimens (**not** the control specimens in 2.5.4) exposed to natural seawater as described in 2.5.3 above shall be cleaned by immersion in the compound at usage concentration for six hours with 5-minutes stirring each half hour. Following cleaning, the test specimens shall be rinsed with water and reimmersed in continuously refreshed, once-through natural seawater for an additional 30 days. After the 30-day seawater exposure, obtain potentiodynamic polarization curves for the test specimens as in 2.5.2.

2.5.6 Pass/fail criteria. For each test specimen, the pass-fail criteria for this test will be that the anodic current density over the range of cupronickel potentials shall meet or exceed that of the control specimens tested in 2.5.3

2.5.7 Seawater properties. Clean, natural seawater with sulfide concentrations less than 0.05 mg/l shall be used. Seawater properties during all portions of the test shall be in accordance with Table II.

TABLE II. Seawater properties.

	TEMPERATURE	SALINITY	pH	DISSOLVED OXYGEN	DISSOLVED OXYGEN	RESISTIVITY
	(°C)	parts per thousand (ppt)		mg/L	Percent Saturation	ohm-cm
MINIMUM	16	35	7.8	4	72	16
MAXIMUM	31	41	8.2	7	89	21

2.6 Identification and markings. Identification and markings shall be permanent and legible and shall include, as a minimum, the manufacturer's name, telephone number and Commercial and Government Entity (CAGE) code in addition to labeling with instructions for use, and appropriate hazard warnings.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.1 Hazardous materials. The compound shall meet all applicable Federal codes and regulations concerning hazardous materials, including package marking.

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:26Z
Check the source to verify that this is the current version before use.

4.2 Market acceptability. The compound shall have been sold previously to the Government or to the commercial market in order to qualify for acceptability under this CID.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Source of documents.

6.1.1 ASTM. ASTM International standards may be obtained from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or online at http://www.astm.org/.

6.1.2 CFR. The Code of Federal Regulations may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20401 or online at www.gpoaccess.gov/index.html.

6.1.3 FAR. The FAR may be obtained from the Regulatory Secretariat, 1800 F Street NW, Washington DC 20405 or online at www.acqnet.gov.

6.1.4 SAE. Society of Automotive Engineers (SAE) Aerospace Materials Specifications (AMS) may be obtained from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001 or online at www.sae.org.

6.1.5 Toxins. The most recent Annual Report on Carcinogens may be obtained from the U.S. Department of Health and Human Services, Public Health Service, National Toxicology Program, P.O. Box 12233, Research Triangle Park, NC 27709 or go to URL http://ntp-server.niehs.nih.gov/ under Public Health, see Report on Carcinogens.

6.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. Product conformance provisions.
c. Quantity required.
d. Packaging requirements.

6.3 Key words.

Acid cleaning
Condenser
Evaporator
Heat exchanger
Seawater system

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:26Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS

Custodians:
Army - AT
Navy - SH
Air Force - 03

Preparing Activity:
Navy - SH
(Project 6850-1469-000)

Review Activities:
Army - EA, MR
Navy - AS
Air Force - 11, 50, 68
DLA - GS, GS3
MISC - MP

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:26Z
Check the source to verify that this is the current version before use.